

1. #include <stdio.h>

```
int main() {
```

```
    int n, i;
```

```
    int bt[20], wt[20], tat[20];
```

```
    printf("Enter number of processes: ");
```

```
    scanf("%d", &n);
```

```
    printf("Enter burst time for each process:\n");
```

```
    for (i = 0; i < n; i++) {
```

```
        printf("P%d: ", i + 1);
```

```
        scanf("%d", &bt[i]);
```

```
    }
```

```
    // Waiting time
```

```
    wt[0] = 0;
```

```
    for (i = 1; i < n; i++) {
```

```
        wt[i] = wt[i - 1] + bt[i - 1];
```

```
    }
```

```
    // Turnaround time
```

```
    for (i = 0; i < n; i++) {
```

```
        tat[i] = wt[i] + bt[i];
```

```
    }
```

```
    printf("\nProcess\tBT\tWT\tTAT\n");
```

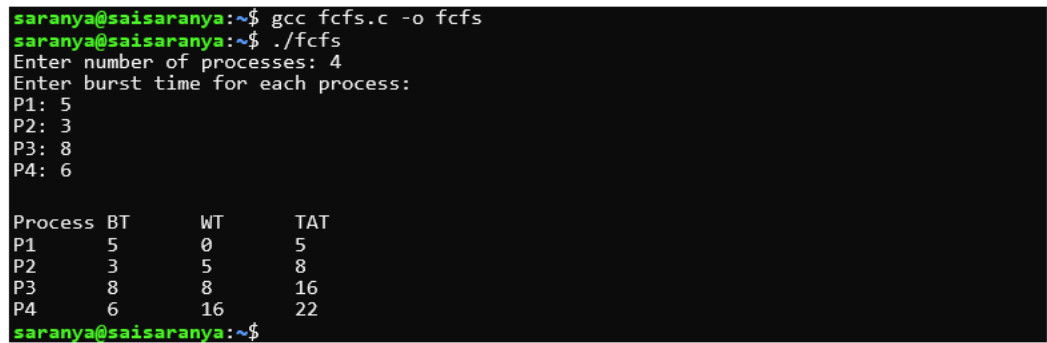
```
    for (i = 0; i < n; i++) {
```

```

        printf("P%d\t%d\t%d\t%d\n",
               i + 1, bt[i], wt[i], tat[i]);
    }

    return 0;
}

```



```

saranya@saisaranya:~$ gcc fcfs.c -o fcfs
saranya@saisaranya:~$ ./fcfs
Enter number of processes: 4
Enter burst time for each process:
P1: 5
P2: 3
P3: 8
P4: 6

Process BT      WT      TAT
P1      5        0        5
P2      3        5        8
P3      8        8       16
P4      6       16       22
saranya@saisaranya:~$

```

2. #include <stdio.h>

```

int main() {
    int n, i, j;

    int bt[20], wt[20], tat[20];

    int temp;

    printf("Enter number of processes: ");
    scanf("%d", &n);

    printf("Enter burst time:\n");

    for (i = 0; i < n; i++) {
        printf("P%d: ", i + 1);
        scanf("%d", &bt[i]);
    }

    // Sort burst times
    for (i = 0; i < n - 1; i++) {

```

```

    for (j = i + 1; j < n; j++) {
        if (bt[i] > bt[j]) {
            temp = bt[i];
            bt[i] = bt[j];
            bt[j] = temp;
        }
    }
}

// Waiting time
wt[0] = 0;

for (i = 1; i < n; i++) {
    wt[i] = wt[i - 1] + bt[i - 1];
}

// Turnaround time
for (i = 0; i < n; i++) {
    tat[i] = wt[i] + bt[i];
}

printf("\nProcess\tBT\tWT\tTAT\n");

for (i = 0; i < n; i++) {
    printf("P%d\t%d\t%d\t%d\n",
        i + 1, bt[i], wt[i], tat[i]);
}

return 0;
}

```

```

saranya@saisaranya:~$ gcc sjf.c -o sjf
saranya@saisaranya:~$ ./sjf
Enter number of processes: 4
Enter burst time:
P1: 6
P2: 8
P3: 7
P4: 3

Process BT      WT      TAT
P1      3        0        3
P2      6        3        9
P3      7        9       16
P4      8       16       24
saranya@saisaranya:~$

```

3. #include <stdio.h>

```

int main() {

    int n, i, j;

    int bt[20], priority[20], wt[20], tat[20];

    int temp;

    printf("Enter number of processes: ");
    scanf("%d", &n);

    printf("Enter burst time and priority:\n");

    for (i = 0; i < n; i++) {
        printf("P%d Burst Time: ", i + 1);
        scanf("%d", &bt[i]);

        printf("P%d Priority: ", i + 1);
        scanf("%d", &priority[i]);
    }

    // Sort according to priority
    for (i = 0; i < n - 1; i++) {
        for (j = i + 1; j < n; j++) {
            if (priority[i] > priority[j]) {
                temp = priority[i];

```

```

        priority[i] = priority[j];
        priority[j] = temp;

        temp = bt[i];
        bt[i] = bt[j];
        bt[j] = temp;
    }
}

// Waiting time
wt[0] = 0;

for (i = 1; i < n; i++) {
    wt[i] = wt[i - 1] + bt[i - 1];
}

// Turnaround time
for (i = 0; i < n; i++) {
    tat[i] = wt[i] + bt[i];
}

printf("\nProcess\tBT\tPriority\tWT\tTAT\n");

for (i = 0; i < n; i++) {
    printf("P%d\t%d\t%d\t\t%d\t%d\n",
        i + 1, bt[i], priority[i], wt[i], tat[i]);
}

return 0;
}

```

```

saranya@saisaranya:~$ gcc priority.c -o priority
saranya@saisaranya:~$ ./priority
Enter number of processes: 4
Enter burst time and priority:
P1 Burst Time: 10
P1 Priority: 3
P2 Burst Time: 5
P2 Priority: 1
P3 Burst Time: 8
P3 Priority: 4
P4 Burst Time: 2
P4 Priority: 2

Process BT    Priority    WT    TAT
P1      10      3         0     5
P2       5      1         5     7
P3       8      4         7    17
P4       2      2        17    25
saranya@saisaranya:~$

```

4. #include <stdio.h>

```
int main() {
```

```
    int n, i;
```

```
    int bt[20], rem_bt[20];
```

```
    int wt[20], tat[20];
```

```
    int quantum;
```

```
    int time = 0;
```

```
    int done;
```

```
    printf("Enter number of processes: ");
```

```
    scanf("%d", &n);
```

```
    printf("Enter burst time:\n");
```

```
    for (i = 0; i < n; i++) {
```

```
        printf("P%d: ", i + 1);
```

```
        scanf("%d", &bt[i]);
```

```
        rem_bt[i] = bt[i];
```

```
        wt[i] = 0;
```

```
    }
```

```
    printf("Enter time quantum: ");
```

```
    scanf("%d", &quantum);
```

```
// Round Robin
```

```

while (1) {
    done = 1;

    for (i = 0; i < n; i++) {
        if (rem_bt[i] > 0) {
            done = 0;

            if (rem_bt[i] > quantum) {
                time = time + quantum;
                rem_bt[i] = rem_bt[i] - quantum;
            }
            else {
                time = time + rem_bt[i];
                wt[i] = time - bt[i];
                rem_bt[i] = 0;
            }
        }
    }

    if (done == 1)
        break;
}

```

// Turnaround time

```

for (i = 0; i < n; i++) {
    tat[i] = wt[i] + bt[i];
}

```

```

printf("\nProcess\tBT\tWT\tTAT\n");

```

```

for (i = 0; i < n; i++) {

```

```
printf("P%d\t%d\t%d\t%d\n",  
      i + 1, bt[i], wt[i], tat[i]);  
}  
  
return 0;  
}
```

```
sarany@saisarany:~$ gcc roundrobin.c -o roundrobin  
sarany@saisarany:~$ ./roundrobin  
Enter number of processes: 3  
Enter burst time:  
P1: 10  
P2: 5  
P3: 8  
Enter time quantum: 2  
  
Process BT   WT   TAT  
P1      10    13    23  
P2       5    10    15  
P3       8    13    21  
sarany@saisarany:~$
```